

Followup 26-07

Chest pain and rapid atrial fibrillation in a 91-year-old

Patient	91-year-old female. Alzheimer's dementia, paroxysmal atrial fibrillation, heart failure with preserved ejection fraction, chronic kidney disease, hypertension, generalized anxiety and depression, chronic leg wound followed by visiting nurses. Home medications included carvedilol, diltiazem, torsemide, aspirin, atorvastatin, quetiapine, and sertraline. Lives alone with daughter support. She had been evaluated in the emergency department the previous day for left arm numbness, weakness, and slurred speech, worked up for stroke, and discharged home.
Outcome	Survived. Admitted as a non-ST elevation myocardial infarction with a troponin of 467 rising to 509, atrial fibrillation with rapid ventricular response, and acute hypoxic respiratory failure. CT angiography ruled out pulmonary embolism. The echocardiogram on hospital day 2 changed the diagnosis: ejection fraction 35 to 40%, down from 66% earlier in the year, with akinesis and ballooning of the mid and apical segments and increased contractility at the base. That pattern is Takotsubo, or stress-induced, cardiomyopathy.
Disposition	Managed medically at the family's election, with no cardiac catheterization. Rate controlled, diuresed, carvedilol changed to metoprolol, diltiazem discontinued once the reduced ejection fraction was known. Discharged home on hospital day 4.

The call

Dispatched for chest pain and difficulty breathing, with a BLS crew already on scene. She was sitting up in a recliner, alert and oriented times four, with substernal chest pain at 6 out of 10, not reproducible with palpation, movement, or respiration. She was breathing at 36 with adequate depth, lungs clear and equal, and she appeared somewhat anxious. She had vomited before arrival.

Her daughter called because of the chest pain and new shortness of breath, and reported that her mother normally has atrial fibrillation. She had been at the hospital the previous day for altered mental status and sent home. The crew reviewed the patient's own chart on scene and found bacteriuria, hyponatremia, anemia, and impaired renal function from that visit, and a discharge heart rate in the 60s.

Partway through the encounter **the pain moved**. She began describing it as upper abdominal rather than substernal, with a soft, nontender, nondistended abdomen. She denied dizziness and syncope.

From contact	BP	MAP	Pulse	Respirations	SpO2
+2 min	148/72	97	156	36	88% (room air)
+31 min	107/50	69	63	36	not recorded

Course, relative to patient contact:

- **+3 min:** Cardiac monitor. **Atrial fibrillation with rapid ventricular response.**

- **+5 min:** Oxygen at 4 L by nasal cannula, which brought the saturation from 88% to 97%.
- **+7 min:** 12 lead ECG. Atrial fibrillation with RVR, **rate varying anywhere between 60 and 180**, no ST segment abnormality.
- **+19 min:** Aspirin 324 mg chewed.
- **+21 min:** IV established.
- **+22 min:** Normal saline 250 mL.
- **+24 min:** Ondansetron 4 mg IV.
- **+25 min:** Nitroglycerin 0.4 mg sublingual.
- **+31 min:** Pain unchanged at 6 out of 10. **Systolic down 41 points.** Further nitroglycerin withheld.

She was less anxious and in no distress by the end of transport, and at points laughed during ordinary conversation.

Two things to hold onto. **The rate was documented as swinging between 60 and 180, and the respiratory rate never moved off 36.**

Primary impression

Chest pain with atrial fibrillation and rapid ventricular response, treated as a possible acute coronary syndrome.

That is the right call and the receiving team made the same one. She was admitted as an NSTEMI, started on heparin, and cardiology was consulted, all on the strength of a troponin of 467 rising to 509 with lateral T wave abnormalities.

Then the echocardiogram on day 2 found an ejection fraction of 35 to 40%, down from 66% in January and 50% in June, with the apex not moving and the base contracting harder to compensate. **That is not what a blocked coronary artery looks like.** A single occluded vessel damages the territory it supplies, and the wall motion abnormality respects that territory. This abnormality crossed the supply of more than one artery and spared the base entirely, which is the signature of Takotsubo cardiomyopathy.

It was never confirmed, and the record is honest about that. The family declined catheterization given her age and dementia, so nobody ever looked at her coronary arteries. The discharge summary carries both the NSTEMI diagnosis and the suspicion of Takotsubo, and says plainly that coronary disease cannot be ruled out.

Hospital course

Emergency department, day 0. Triage found her at 137 with respirations of 26, blood pressure 110/69, and a saturation of 93%. Given atrial fibrillation without anticoagulation, new hypoxia, and chest pain, the differential was pulmonary embolism versus acute coronary syndrome versus heart failure. Diltiazem 5 mg IV was given for rate control along with lactated Ringer's.

Workup showed a white count of 15.2, a lactate of 3.4, an anion gap of 17, a D-dimer of 2,380, and a **troponin of 467, flagged critical.** CT angiography found no pulmonary embolism, but did show bilateral pleural effusions, right worse than left, and a large hiatal hernia. Blood cultures were drawn and ceftriaxone given for the leukocytosis and hypoxia. The two hour troponin came back at 509 and she was started on heparin and admitted for NSTEMI, with the note recording that this might be demand ischemia from the rapid atrial fibrillation.

Day 1. Digoxin was loaded for rate control because her blood pressures were too soft for more beta blockade. Carvedilol was replaced with metoprolol. Intravenous furosemide was started for the effusions. Cardiology saw

her and documented that the family was clear that invasive testing was not in line with goals of care. Troponin peaked at 509 and began falling.

Day 2. The echocardiogram was done. Ejection fraction 35 to 40%, mid to apical akinesis and ballooning with hypercontractile basal segments, reduced right ventricular apical function with preserved base, and no thrombus. The report named biventricular Takotsubo as the consideration. **Diltiazem was discontinued on the strength of that ejection fraction**, since it depresses contractility in a ventricle that could not afford it. Metoprolol was increased, and an SGLT2 inhibitor and losartan were started for heart failure.

Days 3 and 4. Rate controlled in the seventies. Intravenous diuresis stopped because creatinine and BUN were climbing, and home torsemide resumed. Heparin completed at 48 hours. Blood cultures never grew anything and the leukocytosis resolved without further antibiotics. **Long term anticoagulation was not started, on the grounds of fall and bleeding risk.** Discharged home.

Final diagnoses: NSTEMI with suspected Takotsubo cardiomyopathy; acute on chronic heart failure with reduced ejection fraction; paroxysmal atrial fibrillation with rapid ventricular response; acute hypoxic respiratory failure; bilateral pleural effusions; leukocytosis, resolved; chronic kidney disease; Alzheimer's dementia.

Education points

New rapid atrial fibrillation is usually a symptom, not the disease

Atrial fibrillation with a rapid ventricular response gets treated as a rhythm problem, and in a patient whose baseline is sinus and who has just gone into it, the rhythm is very often the body's response to something else.

The mechanism is straightforward. Atrial fibrillation happens when the atria depolarize chaotically instead of in an organized way, and the ventricle responds to whatever fraction of those impulses gets through the AV node. **The AV node is under autonomic control.** Sympathetic drive speeds its conduction and lets more impulses through. So anything that produces a catecholamine surge, which includes pain, hypoxia, fever, infection, hypovolemia, and fear, will drive the ventricular rate up in a patient who has the substrate for atrial fibrillation.

That is why this patient's rate is the most interesting number in the chart. She was discharged the previous day with a rate in the sixties. On this call it swung between 60 and 180, hit 156 on the crew's first set, and was back at 137 at triage nineteen minutes later. **Nobody gave her anything to slow it down between those readings.** The rate was not a stable disease state. It was a variable tracking her physiology.

And she had four separate catecholamine drivers running at once: chest pain at 6 out of 10, hypoxia at 88%, an acute cardiomyopathy, and the anxiety the crew documented. Rate control in the field is not the priority in that situation, and the record shows why. She got one dose of diltiazem in the emergency department, then needed digoxin because her pressure could not tolerate more, and the drug she had been on at home was eventually stopped outright once her ejection fraction was known. **Treat what is driving the rate, and be careful about the rate itself.**

Why the pressure dropped 41 points

The narrative attributes the drop to the nitroglycerin, which is the obvious reading. The honest version is more useful, because three separate things were reducing her stroke volume at the same time.

She lost her atrial kick. In sinus rhythm, atrial contraction at the end of diastole adds a final push of blood into the ventricle, contributing roughly 20 to 30% of the volume it will eject. In atrial fibrillation the atria quiver instead of contracting and that contribution is gone. It matters more in a stiff ventricle, and hers was stiff, since she carries a diagnosis of heart failure with preserved ejection fraction.

Her rate of 156 removed filling time. Diastole is when the ventricle fills, and it is also the phase that shortens first as rate climbs. At 156 there is very little of it left. Less filling time means a smaller volume in the ventricle at the end of diastole, which means a smaller stroke volume, and cardiac output is stroke volume times rate. Past a certain rate, going faster stops helping and starts hurting.

Nitroglycerin took her preload. At the sublingual dose the dominant effect is venodilation, which pools blood in the venous system and returns less of it to the heart. That is the intended effect when you are trying to unload a congested ventricle. In a patient who had eaten and drunk poorly for some time, with dry mucous membranes documented in the emergency department and a BUN to creatinine ratio of 28, preload was part of what was holding her output up.

Three hits on the same variable, and the pressure fell. The crew saw it, held further nitroglycerin, and documented the reason, which is the correct sequence.

One more piece of honesty about that number. The 41 point comparison spans 29 minutes, and inside that window she also received 250 mL of fluid, ondansetron, and a ventricular rate that fell from 156 to 63 on its own. **The chart cannot isolate what caused what**, and 19 minutes after that reading her systolic was back to 110 with a rate of 137. That is not a criticism of anybody's documentation. It is what a moving target looks like, and it is worth recognizing when you are tempted to attribute a change to the last thing you did.

Takotsubo cardiomyopathy

The apex of the left ventricle stopped moving and ballooned outward, while the base contracted harder than normal. On a ventriculogram that shape resembles a Japanese octopus trap, a pot with a round body and a narrow neck, which is where the name comes from.

The current understanding is that it is catecholamine mediated. A surge of adrenaline, from emotional or physical stress, stuns myocardium directly rather than through a blocked artery. The apex appears to be more susceptible than the base, and the result is transient systolic failure that looks exactly like a large anterior myocardial infarction: chest pain, ECG changes, and a troponin rise.

It is not distinguishable from an infarction without imaging. She had chest pain, lateral T wave changes, a critical troponin, a HEART score of 8, and every reason to be treated as an NSTEMI, which is what happened and what should have happened. The two findings that separated it were an echocardiogram showing wall motion abnormality crossing more than one coronary territory, and an ejection fraction that had fallen from 66% to 35 to 40% in eight months.

Two things worth carrying. It is **typically reversible**, with function often recovering over weeks, which is a very different prognosis from infarcted muscle. And it **predominantly affects older women**, which is exactly this patient.

Further reading on Takotsubo. The variants, the diagnostic criteria, and the complications go well past what a case document can carry. StatPearls has a free chapter: <https://www.ncbi.nlm.nih.gov/books/NBK430798/>

Prehospital considerations

A respiratory rate that will not come down is telling you something. Hers was 36 at first contact and 36 half an hour later, unchanged while her heart rate fell 93 points and her oxygen saturation was corrected to 97%. A respiratory rate that does not respond to oxygen is not being driven by oxygen. Hers was being driven by a lactate of 3.4 and an anion gap of 17, blowing off CO₂ to buffer an acid load, and by pleural effusions. Fix the hypoxia, and then ask what is still making her breathe that fast.

A return visit inside 24 hours is a finding in its own right. She was discharged the previous day and called again the next. Whatever was or was not found the first time, the fact of the return raises the pretest probability of everything, and the receiving team needs to hear it in the first sentence of your report. The crew had that detail and it appears in every hospital note downstream.

In dementia, the earliest history is often the most reliable. Four different accounts of this patient's chest pain exist across the record, differing in location, timing, and whether it was still present. Hers is a diagnosis that erodes recall, and each retelling was further from the event. The crew's account, taken while the pain was still happening, is the closest thing to a primary source that this chart contains, including the detail that the pain moved from substernal to upper abdominal.

Chart review on scene is worth doing and worth double checking. Pulling up the previous visit is not something most crews think to do, and it gave this crew four data points the receiving team would otherwise have had to look up. Three of them match the record exactly. The white count does not: the previous day's was 7.5, not 1. It is worth knowing what the wrong version would have cost, because a white count of 1 in a 91-year-old with bacteria in her urine is neutropenic sepsis until proven otherwise, and it points a receiving team somewhere the patient was not. When you relay a number you read rather than measured, say where it came from.

The interventions

PREHOSPITAL INTERVENTIONS

Oxygen at 4 L, 88% to 97%. The single most valuable thing done on this call. She was hypoxic on room air with a failing ventricle and a rate of 156, and hypoxia is itself one of the drivers pushing that rate. Correcting it treats the arrhythmia indirectly and buys the myocardium oxygen it was not getting.

Cardiac monitor and 12 lead. Both, early, on a chest pain patient. The 12 lead documented atrial fibrillation with RVR and no ST elevation, which is what let the receiving team move straight to a non-ST elevation pathway. The observation that the rate swung between 60 and 180 is not something a single strip captures and it is not something the hospital could reconstruct later.

Aspirin 324 mg. Irreversibly inhibits cyclooxygenase in platelets, reducing thromboxane A2 and platelet aggregation for the life of the platelet. It is the intervention with the best mortality evidence in acute coronary syndrome and it costs nothing. Correct here even though the final diagnosis turned out to be a stress cardiomyopathy, because nobody could have known that, and she was continued on aspirin through the admission and at discharge.

Normal saline 250 mL. A modest, sensible volume in a patient with heart failure and a history of pleural effusions. Small enough not to flood her, enough to support preload in a patient who had been eating and drinking poorly.

Ondansetron 4 mg IV. Blocks serotonin at the 5-HT3 receptor in the chemoreceptor trigger zone and the gut. Reasonable for a patient who had vomited. One thing to know about it in this specific patient: ondansetron prolongs the QT interval, and while that rarely matters in isolation, it is a consideration in a patient with an unstable rate and electrolytes you cannot see. Nothing came of it here.

Nitroglycerin 0.4 mg sublingual, then withheld. Discussed above. The important part is the second half. The pressure dropped, the pain did not improve, and no further nitroglycerin was given, with the reason documented. **Recognizing that a drug did not work and stopping is a clinical decision, and it should be charted as one.**

IN-HOSPITAL INTERVENTIONS

Diltiazem 5 mg IV, and why it was later stopped. Diltiazem is a non-dihydropyridine calcium channel blocker. It slows conduction through the AV node, which is how it controls rate in atrial fibrillation, and it also reduces the force of ventricular contraction. That second effect is the problem. When the echocardiogram found an ejection fraction of 35 to 40%, cardiology discontinued it, because a negative inotrope in a ventricle that is already failing makes the failure worse. **The right rate control drug depends on the ejection fraction, and nobody knew hers until day 2.** That is part of why aggressive rate control is not a field priority in an undifferentiated patient.

Digoxin load. Chosen on day 1 specifically because her blood pressures were soft. Digoxin slows the ventricular rate through vagal effects on the AV node while mildly increasing contractility, which makes it one of the few rate control options that does not drop blood pressure. It is slow, and it is a narrow therapeutic window drug in a patient with kidney disease, which is why it was held once the rate settled.

Heparin infusion for 48 hours. Standard for a non-ST elevation acute coronary syndrome being managed medically, preventing clot propagation while the myocardium is at risk.

CT angiography of the chest. Ordered on a D-dimer of 2,380 to exclude pulmonary embolism, which was the correct concern in a patient with atrial fibrillation not on anticoagulation, new hypoxia, and chest pain. Negative for embolism, and it found the pleural effusions that explained part of her hypoxia.

The anticoagulation decision. She has atrial fibrillation, is not on an anticoagulant, and was not started on one. That looks like an omission and it is a deliberate judgment: her stroke risk from atrial fibrillation was weighed against a documented fall history and bleeding risk at 91 with dementia, and the team decided against. **Worth knowing when you find an atrial fibrillation patient with no anticoagulant on the list, because it is frequently a decision rather than a gap.**

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Lab	Day before	Day 0	Day 2	Reference range
hs troponin T (ng/L)	not drawn	467, then 509	361	Less than 15 (female)
Lactic acid (mmol/L)	not drawn	3.4, then 2.6	not repeated	0.0 to 2.0
Anion gap (mmol/L)	not reported	17	12	5 to 14
Bicarbonate (mmol/L)	26	21	22	22 to 29
D-dimer (ng/mL FEU)	not drawn	2,380	not repeated	0 to 500
White blood cell ($\times 10^9/L$)	7.5	15.2	11.2	4.0 to 10.8
Neutrophils, absolute ($\times 10^9/L$)	not reported	13.47	not reported	1.80 to 7.80
Lymphocytes, absolute ($\times 10^9/L$)	not reported	0.70	not reported	1.00 to 4.00
Hemoglobin (g/dL)	10.5	10.3	11.0	12.0 to 16.0
BUN (mg/dL)	38	34	36	8 to 23
Creatinine (mg/dL)	1.45	1.21	1.32	0.50 to 1.00

Lab	Day before	Day 0	Day 2	Reference range
AST (U/L)	23	56	67	0 to 35
Total bilirubin (mg/dL)	not reported	1.6	0.9	0.2 to 1.3

The troponin is the number that drove every decision, and it could not answer the question everyone was asking. It went 467, then 509, then began falling. That rise and fall is the pattern of acute myocardial injury and it is why she was heparinized and admitted.

Troponin is a protein of the cardiac contractile apparatus. It appears in blood when heart muscle cells are damaged, and the high sensitivity assay detects it at very low concentrations, which is why the female cutoff is 15 rather than 150. **What it tells you is that myocardium was injured. It cannot tell you why.** A blocked coronary artery does it. So does demand outstripping supply in a heart running at 156 while the patient is hypoxic. So does a catecholamine surge stunning the apex. All three were on this patient's differential, the treating team named the second explicitly, and the imaging eventually pointed at the third.

The lactate and the anion gap explain the respiratory rate of 36. Lactate 3.4 with an anion gap of 17 and a bicarbonate of 21 is a metabolic acidosis with an unmeasured anion filling the gap. The body's fastest response to acid is to blow off CO₂, and that is done by breathing faster. Her rate of 36 was compensation, not distress, and it is why oxygen corrected her saturation without touching her respiratory rate. The lactate came down to 2.6 on recheck and the gap closed to 12 by day 2 as her rate came under control and her perfusion improved.

The white count is the pattern, not the number. It went from 7.5 the day before to 15.2, then back to normal within two days on essentially no antibiotic treatment, with negative blood cultures and a chest CT showing no pneumonia. Look at which lines moved: **neutrophils up at 13.47 and lymphocytes down at 0.70.** That combination is the stress response, where cortisol pushes neutrophils out of the marrow and simultaneously pulls lymphocytes out of the circulation. It is also what bacterial infection produces, which is exactly why a white count in an acutely stressed patient cannot diagnose infection by itself. The treating team called it unclear significance and possibly reactive, and it resolved on its own.

The D-dimer is a rule-out test with almost no specificity here. D-dimer is a fragment released when the body breaks down a clot, so it rises whenever clot is being formed and dissolved anywhere. For patients over 50 an age adjusted cutoff of age times ten is accepted, which for her would be 910 rather than 500. Hers was 2,380, positive either way. **The point is not that her result was high, it is that almost nothing would have made it low.** The lab's own comment lists renal failure, severe infection, and cancer among the conditions where D-dimer should not be used to rule out embolism, and she has chronic kidney disease. A negative D-dimer would have been useful. A positive one in a 91-year-old with kidney disease means you are getting a CT.

Two values worth reading on your own. Her **AST rose from 23 to 56 and then 67 while her ALT stayed nearly normal**, which is not a hepatocellular pattern. AST is abundant in cardiac and skeletal muscle as well as liver, and a rising AST with a flat ALT in a patient with an acute cardiomyopathy is muscle, not liver. Her **total bilirubin of 1.6 with a normal alkaline phosphatase** is a mild, isolated elevation that resolved to 0.9, which is the pattern of a congested liver in acute right sided heart failure, and her echocardiogram found reduced right ventricular apical function.

Learning points

1. **New rapid atrial fibrillation in a sick patient is usually a symptom of something else.** The AV node is under autonomic control, so pain, hypoxia, fever, infection, and hypovolemia all push the ventricular rate up in

someone with the substrate for it. This patient's rate swung between 60 and 180 with nothing given to change it, and was back at 137 nineteen minutes after being 63. Find and fix the driver before chasing the number.

2. **Three things were taking her stroke volume at once.** Atrial fibrillation cost her the atrial kick, which is 20 to 30% of ventricular filling. A rate of 156 removed the diastolic time she needed to fill. Nitroglycerin took her preload on top of both. The pressure fell 41 points, the pain did not change, further nitroglycerin was correctly withheld, and the reason was documented.
3. **A troponin tells you myocardium was injured, not why.** A blocked artery, demand outstripping supply at a rate of 156, and a catecholamine surge stunning the apex all produce the same rising and falling curve. This patient was treated as an infarction, which was right, and the echocardiogram two days later found wall motion abnormality crossing more than one coronary territory, which is Takotsubo and not a blocked artery.
4. **A respiratory rate that does not respond to oxygen is not a breathing problem.** Hers stayed at 36 while her saturation corrected to 97% and her heart rate fell 93 points. She was compensating for a lactate of 3.4 and an anion gap of 17. Correct the hypoxia, then ask what is still driving the work.
5. **A return visit within 24 hours is itself a finding.** She was discharged the previous day and called again the next morning. Whatever was found or missed the first time, the return raises the pretest probability of everything and belongs in the first sentence of your report. In a patient with dementia, your history is also the earliest and usually the most reliable one anybody will get.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>